MILITARY SPECIFICATION

BUCKLES, END, 1-INCH

This specification is approved for use by all Departments and Agencies o the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and size of 1-inch end buckle

1.2 Classification. Buckles shall be of the following classes as specified (see 6.2).

Class 1 - Brass.
Class 3 - Malleable iron.
Class 4 - Steel.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-613 - Brass, Leaded and Nonleaded: Flat Products (Plat Bar, Sheet, and Strip)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5340

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-16T13:53Z
Check the source to verify that this is the current version before use.

TT-E-529 - Enamel, Alkyd, Semigloss
PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-665 - Boxes: Paperboard, Metal Edged and Components
PPP-B-676 - Boxes, Setup

MILITARY

MIL-F-495 - Finish, Chemical, Black, For Copper Alloys

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

4-1-157 - Buckles, End, 1"

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 47 - Standard Specification For Malleable Iron Castings
A 366 - Steel, Carbon, Cold Rolled Sheet, Commercial Quality
B 633 - Electrodeposited Coatings of Zinc on Iron and Steel
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Source: https://assist.dla.mil -- Downloaded: 202640-16T13:53Z
Check the source to verify that this is the current version before use.

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

* 3.1 Materials. Materials shall comply with applicable document and requirements specified herein. Recycled material may be used (see 6.3).

* 3.1.1 Brass sheet or strip. The brass sheet or strip shall conform to copper alloy number 268 of QQ-B-613, quarter-hard or half-hard as specified herein.

* 3.1.2 Malleable iron. The malleable iron shall conform to ASTM A 47 except that the requirements for marking the castings are not applicable.

* 3.1.3 Steel strip. The steel strip shall be cold rolled as specified in ASTM A 366.

* 3.1.4 Enamel. The enamel shall conform to type I or II of TT-E-529. The color shall be black.

3.2 Construction. The construction of the buckles shall conform to the requirements specified herein and on Drawing 4-1-157.

3.2.1 Class 1 buckles.

* 3.2.1.1 Body. The body for class 1 buckles shall be nominal 0.0808 inch in thickness and stamped from brass sheet or strip, copper alloy number 268, quarter-hard, as specified in 3.1.1.

* 3.2.1.2 Clip. The clip for class 1 buckles shall be nominal 0.0253 inch in thickness and stamped from brass sheet or strip, copper alloy number 268, half-hard, as specified in 3.1.1.

3.2.2 Class 3 buckles.

3.2.2.1 Body. The body for class 3 buckles shall be cast from malleable iron as specified in 3.1.2 and Drawing 4-1-157.

Source: https://assist.dla.mil -- Downloaded: 202610-16T13:53Z
Check the source to verify that this is the current version before use.

3.2.2.2 Clip. The clip for class 3 buckles shall be nominal 0.0239 inch in thickness and stamped from steel strip as specified in 3.1.3.

3.2.3 Class 4 buckles.

3.2.3.1 Body. The body for class 4 buckles shall be nominal 0.0747 inch in thickness and stamped from steel strip as specified in 3.1.3.

3.2.3.2 Clip. The clip for class 4 buckles shall be nominal 0.0239 inch in thickness and stamped from steel strip as specified in 3.1.3.

3.3 Finish. Prior to application of a finish, all burrs and sharp edges shall be removed from stampings. Castings shall be smooth, sound, free from pits, blowholes, porosity, fractures and cracks.

3.3.1 Class 1 brass buckle. Class 1 brass buckles shall be given a black chemical finish in accordance with MIL-F-495, except that the gloss requirements shall be 40 maximum.

* 3.3.2 Class 3 malleable iron and class 4 steel buckles. Class 3 malleable iron and class 4 steel buckles shall be given a phosphate treated zinc plate conforming to type IV, classification Fe/Zn Sc3 of ASTM B 633, followed by bake enameling. The enamel shall be as specified in 3.1.4. The coating shall level out to a uniform dry film thickness without objectional orange peel, wrinkles, drops, streaks or areas of no film.

3.4 Marking for identification. All buckles shall be marked for identification in accordance with MIL-STD-130. The letters "US" and manufacturer's identification of such known character as to be easily identifiable shall be permanently affixed by stamping or etching on the back of the buckle. The identification marks shall be incorporated in such a manner that the surface opposite those markings shall show no signs of penetration or unevenness or cause distortion in any part of the buckle.

3.5 Workmanship. The finished buckles shall conform to the quality and grade of product established by this document.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection

Source: https://assist.dla.mil -- Downloaded: 20261O-16T13:53Z
Check the source to verify that this is the current version before use.

requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 Intermediate inspection. Inspection shall be conducted to see that class 3 and class 4 buckles are given a phosphate-treated zinc plate as specified in 3.3.2 are in compliance with ASTM B 633.

4.2.3 End item inspection. The buckles shall be examined for visual defects and proper dimensions. The lot shall be all the buckles of one class offered for inspection at one time. The sample unit for this examination shall be one assembled buckle.

4.2.3.1 Visual examination. Examination shall be made of the buckles for the defects in table I. The inspection level shall be II, and the acceptable quality level (AQL) shall be 4.0 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish:			
Black enamel (classes 3 and 4)	Color not as specified	101	
	Area of no film		201
	Finish not uniform, wrinkles, orange peel, drops, or streaks		202

Source: https://assist.dla.mil -- Downloaded: 202610-16T13:53Z
Check the source to verify that this is the currentversion before use.

TABLE I. Classification of defects - Continued

Examine	Defect	Classification	
		Major	Minor
Black chemical (class 1)	Color not as specified, or finish not clean or uniform		203
	Dirt, smudge, oil, grease, or other foreign matter on surface		204
	Any evidence of corrosion under finish	102	
	Area of no finish or finish discolored		205
Construction and workmanship (all classes as applicable)	Unsound casting, pits, blowholes, porosity, fractures, or cracks	103	
	Stampings distorted	104	
	Mutilation such as metal mark or dig, or gouges clearly noticeable		206
	Rough or sharp edges (e.g., fin, flash, sprue, or gate)	105	
Assembly	Clip not securely assembled or becomes easily detached	106	
Identification marking	Missing, incomplete, illegible, or incorrect		207
	Indication of penetration, unevenness or distortion; marking on wrong side of buckle		208

4.2.3.2 Dimensional examination. Inspection shall be made of buckles for compliance with specified dimensions. Any deviation shall constitute a defect. The inspection level shall be S-2, and the AQL shall be 4.0, expressed in terms of defects per hundred units.

4.2.3.3 Testing of class 1 end item. Testing of a completely fabricated class 1 buckle shall be performed in accordance with MIL-F-495 except that the AQL shall be 6.5, expressed in terms of defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:53Z
Check the source to verify that this is the current version before use.

4.2.4 Packaging inspection. An examination shall be made to determine compliance with the applicable preservation, packing, and marking requirements of section 5 of this document. Defects shall be scored in accordance with table II. The lot shall be the number of shipping containers offered for inspection at one time. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2, and the AQL shall be 2.5, expressed in terms of defects per hundred units.

TABLE II. Packaging inspection

Examine	Defect
Markings (exterior and interior)	Omitted, incorrect, illegible, or of improper size, location, sequence, or method of application
Materials	Any component missing or damaged
Contents (exterior and interior container)	Number per container is more or less than required

* 4.2.5 Palletization inspection. When palletization is specified (see 6.2), an examination shall be made to determine compliance with the applicable requirements. Characteristics to be examined and defects applicable thereto shall be as specified in table III. The lot size shall be the number of palletized unit loads in the inspection lot. The sample unit shall be one palletized unit load fully packaged. The inspection level shall be S-1, and the AQL shall be 6.5, expressed in terms of defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:53Z
Check the source to verify that this is the current version before use.

TABLE III. Examination of palletization

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement
Palletization	Not as specified Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted, incorrect, illegible, or of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

* 5.1.1 Level A. Unless otherwise specified (see 6.2), eight hundred and sixty buckles of one class only, shall be unit packed in a snug-fitting, fiberboard box conforming to style RSC, type CF or SF, class domestic, grade 200 of PPP-B-636. Each box shall be agitated from time to time to ensure a compact and well-filled box. Each box shall be closed in accordance with the appendix of the box document. When specified (see 6.2), alternative boxes complying with PPP-B-566, PPP-B-665, PPP-B-676, or PPP-B-636 may be used. The type of box may be selected by the contractor except that weight and dimensional limitations shall not be exceeded.

5.1.2 Commercial. Buckles shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Unless otherwise specified (see 6.2), five thousand one hundred sixty buckles, preserved as specified in 5.1, shall be packed in a snug-fitting, fiberboard shipping container conforming to style FTC, grade V2s

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:53Z
Check the source to verify that this is the current version before use.

of PPP-B-636, except that the weight limitations shall be waived. Each shipping container shall be closed and reinforced with strapping or tape banding in accordance with the appendix of the container document.

* 5.2.2 Level B packing. Unless otherwise specified (see 6.2), five thousand one hundred sixty buckles, preserved as specified in 5.1, shall be packed in a snug-fitting, fiberboard shipping container conforming to style FTC, type CF, class domestic, variety DW, grade 350 of PPP-B-636. Each shipping container shall be closed and reinforced as specified in 5.2.1.

5.2.2.1 Weather-resistant containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636.

* 5.2.3 Commercial packing. Buckles, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), link assemblies, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size that does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The buckles are primarily designed for use with 1-inch webb straps for equipage items.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Class of buckle required (see 1.2).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 202640-16T13:53Z
Check the source to verify that this is the currentversion before use.

d. Whether quantity per unit pack, type of container, and pack is other than specified (see 5.1.1, 5.2.1, and 5.2.2).
e. Whether weather-resistant class fiberboard shipping containers are required for level B packing (see 5.2.2.1).
f. When palletization is required (see 5.3).

* 6.3 Recycled material. The contractor is encouraged to use recycled material when practical, provided that it meets the requirements of this document (see 3.1).

6.4 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue have been made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 82

Review activity:

DLA - IS

Preparing activity:

Army - GL

Project No. 5340-1632

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:53Z
Check the source to verify that this is the current version before use.

TRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in mitting comments and suggestions for improvements. All users of military standardization documents are invited to provide gestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and led. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the blems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an nowledgement will be mailed to you within 30 days to let you know that your comments were received and are being sidered.

TE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of cification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

ARTMENT OF THE ARMY
Natick Research and Development Center
Natick, Massachusetts 01760

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

ICIAL BUSINESS
ALTY FOR PRIVATE USE $300
STRNC-ES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research and Development Center
ATTN: STRNC-ES
Natick, MA 01760

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:53Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-1962C	Buckles, End, 1-Inch

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:53Z
Check the source to verify that this is the current version before use.